

Insert a Duvet into Its Cover SOP (1x Episode: one duvet, one cover, in situ)

One episode puts one duvet all the way inside one cover on the bed where the bed stands. The duvet is opened out flat. Its two head corners are tucked into the two closed corners of the gathered cover. The cover is then unrolled down the duvet to the foot end. The foot corners are pushed in, the opening is shaken once to seat the duvet, the buttons are closed, and the cover is spread flat on the bed. The episode ends when the duvet sits square inside the closed cover and the cover lies flat and even on the mattress. Never split the filling and the spreading into two episodes.

The base is **passive**: it has no drive of its own, so it is pushed by hand up to the long side of the bed and locked there, and nothing is carried away to a table. Everything the episode touches is already on the bed when recording starts: the mattress with its fitted sheet on, the folded duvet, and the cover gathered into a band at the head end.

The bed is worked **as found**. The mattress has its fitted sheet on and nothing else on top. The **headboard stands at the left end** of the bed as the arms see it and the **foot is at the right end**, so the **left half** of the mattress is the head half and the **right half** is the foot half.

This is an in-situ task, and four things follow from that. First, **all the work happens on the mattress**: the duvet and the cover are never lifted up into the air and held there. The fabric stays on the mattress and slides along it. Second, **the work runs from the head end to the foot end** and never the other way: the two head corners are tucked first and the cover unrolls from them toward the foot. Third, **the bed never moves**: nothing leans on, pushes, drags, or lifts the bed frame, the headboard, or the mattress. Fourth, **nothing goes on the floor**: fabric that touches the floor is dirty, it is not picked up, and it is reported.

The mattress is set up in one of three ways. The bed, the headboard, and the gathered cover never move. Only the folded duvet does.

- **Config L**: the folded duvet lies in the **head half**, against the band, with its free edges toward the foot.
- **Config M**: the folded duvet lies **centered on the mattress**, with its folded edge toward the head and its free edges toward the foot.
- **Config R**: the folded duvet lies in the **foot half**, with its folded edge at the foot and its free edges toward the head.

One config per episode, chosen before recording and never changed mid-episode. Where a step depends on the setup it says so on an **IF** line. Look at the bed and follow the line that matches.

Same-side rule: the gripper on the side the free edge of the folded duvet lies nearer is the one that draws the duvet open, and the gripper on a corner's side is the one that tucks it. After that the roles are fixed: the **left gripper** owns the head end, where both head corners are tucked, and the **right gripper** owns the foot end, where the foot corners are pushed in, the opening is shaken, and the buttons are closed. The cover is unrolled down the duvet by whichever gripper the band is nearer, while the other gripper pins the duvet still.

The order never changes: **open the duvet flat; tuck the two head corners; unroll the cover to the foot; push the foot corners in; shake the opening once; close the buttons; spread the cover on the bed.**

Setup

Complete the base positioning and all the checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the near long side of the bed, with the **headboard to the left** and the **foot to the right**. Stop it **square to the side edge of the mattress**, so the edge runs straight across the frame and neither end of the mattress sits nearer than the other.
2. Stop it **centered on the long side**, so the head end and the foot end of the mattress are the same distance from the middle of the base.
3. Stop it **close enough** that either gripper reaches the far edge of the mattress at the middle without the arm extending, and **far enough** that no arm, wrist, or part of the base touches the bed frame or the mattress edge while both arms work.
4. Check the **head end**: with the base parked, the **left gripper** reaches the band against the headboard, the front head corner of the mattress, and the back head corner of the mattress, and closes on each without extending.
5. Check the **foot end**: the **right gripper** reaches the front foot corner and the back foot corner of the mattress, and the whole foot edge between them where the buttons will lie, without extending.
6. Check the **sweep**: each gripper slides along its own half of the mattress from the middle to its end without hitting the other arm and without hitting the headboard.
7. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.

8. If any of lines 1 to 6 fails, push the base to a new park by hand and start again at line 1. Do not work a bed the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the bed or the headboard hard enough to shift it, nothing and nobody touches it, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the middle of the mattress and its frame includes the whole mattress, the head end with the band, the foot end, and the folded duvet.
3. The camera reads the mattress from the side, so where the fabric lies and how far it reaches are readable.
4. The camera reads both head corners of the mattress, so a tucked corner and a corner that has slipped out can be told apart.
5. The camera reads the opening of the cover and its buttons at the foot end, so open buttons and closed buttons can be told apart.
6. The camera reads the top face of the cover at the end, so whether it lies flat and even is readable.
7. Both arms are at home with grippers open.
8. The **left arm** reaches the band, both head corners of the mattress, and the whole head half of the mattress, without extending to a joint limit. It never needs to reach the foot half.
9. The **right arm** reaches both foot corners of the mattress, the foot edge, and the whole foot half of the mattress, without extending to a joint limit. It never needs to reach the head half.
10. Both arms slide fabric along the mattress at mattress height. Neither arm lifts a load above the headboard.
11. The two arms do not collide. An arm pinning the duvet at the middle stays out of the other arm's path along the bed.
12. Nothing hangs off the arms that can catch on the fabric.
13. If a place cannot be reached, re-park the base by the Base positioning steps until lines 8 to 12 hold.

Materials checklist

1. The **bed** is one mattress on one frame with a **headboard at the head end**. Nothing moves it, leans on it, or sits on it.
2. The mattress has a **fitted sheet** on it. The sheet is smooth and tucked at every corner. Nothing else lies on the mattress: no pillow, no blanket, no clothes.
3. The **duvet** is one duvet that fits the cover. It is dry, clean, and has no tears. It has no cover on it at the start.

4. The duvet lies **folded in half once** across its length, so it is half as long as the mattress and its full width, at the spot this episode's config puts it: the **head half against the band in Config L, centered in Config M, the foot half in Config R**. Its free edges point toward the foot in Config L and M and toward the head in Config R. No part of it hangs on the floor.
5. The **cover** is one duvet cover that fits the duvet. It is dry, clean, right side out, and closes with **buttons** along its opening. Every button is open at the start. No button is missing or loose, and no button hole is torn.
6. The cover is **gathered into a band**: by hand, before recording, it is bunched from its opening back toward its closed end, like a sock gathered to its toe, until it is a band no wider than a hand. The band lies across the head end of the mattress, pushed against the headboard, running from the front edge to the back edge, with its two closed corners at the front and the back and their **pockets facing the foot**. The opening and the buttons are inside the band.
7. The mattress between the band and the folded duvet, and everywhere else, is clear.
8. The floor along the near side of the bed, where the base parks, is clear and dry. Nothing hangs over the side of the bed at the start.
9. The room floor is clean and dry, so fabric that touched it can be told from fabric that did not.

Workspace layout

Nothing is marked or taped out. Every place below is judged by eye from the bed itself.

- **Mattress**: the whole work surface. All the work happens on it.
 - **Head half**: the left half, toward the headboard. The left gripper's side.
 - **Foot half**: the right half, toward the foot. The right gripper's side.
- **Head end**: the end of the mattress at the headboard. The band lies here, and both head corners are tucked here.
- **Foot end**: the end of the mattress away from the headboard. The opening and the buttons end up here.
- **Band**: the gathered cover, across the head end against the headboard, pockets facing the foot. It is unrolled down the mattress and ends up spread over the whole duvet.
- **Duvet spot**: where the folded duvet lies at the start: head half in Config L, centered in Config M, foot half in Config R. It empties as the duvet is drawn open.
- **Bed frame and headboard**: never moved, never leaned on, never used to hold fabric.
- **Floor**: out of bounds. No fabric ever touches it.

Arm assignments

- **The left gripper owns the head end**. It tucks both head corners of the duvet into the pockets of the band, the front corner first and then the back corner. It starts the unrolling

when the band is in the head half, and it pins the duvet at the head half once the band has crossed the middle.

- **The right gripper owns the foot end.** It pins the duvet at the middle while the band is unrolled through the head half, takes over the unrolling once the band crosses the middle, pushes the foot corners in, shakes the opening, and closes every button.
- **The gripper on the free edge's side opens the duvet:** the right gripper in Config L and M, the left gripper in Config R.
- **Spreading is shared:** the left gripper squares the head end, the right gripper evens the two long edges from the middle, and each gripper smooths its own half.
- Fabric is only ever pinched at an edge or a corner, or pinned flat with a closed gripper. Nothing is gathered up in a gripper and lifted.

Vocabulary

- **In situ:** the bed is worked where it stands. Nothing is carried away to a table, and nothing about the room is rearranged for the task.
- **Passive base:** the base has no drive. It is pushed by hand to its park before recording, locked there, and it does not move at all during the episode.
- **Config:** which of the three duvet spots this episode uses. It is set before recording and never changes mid-episode.
- **Band:** the cover gathered by hand from its opening back to its closed end, lying as a narrow band across the head end of the mattress with its pockets facing the foot.
- **Pocket:** one of the two closed corners of the cover, standing open at the front or the back of the band. A duvet corner goes into it. **Unvalidated:** whether a tucked corner stays in its pocket once the gripper opens is to be checked at the station.
- **Head corners:** the two corners of the duvet nearest the headboard once the duvet is open, the front head corner and the back head corner. These go into the pockets.
- **Foot corners:** the two corners of the duvet nearest the foot. These are pushed into the cover's near corners inside the opening.
- **Tucked:** the duvet corner sits right inside its pocket, tip into tip, with no slack between them.
- **Pin:** a closed gripper pressed flat on the duvet, hard enough that the duvet does not slide when the cover is pulled, and no harder. A pin never lifts or moves the duvet.
- **Unroll:** the band is pinched at its middle and drawn toward the foot a hand's width at a time, so the gathered cover pays out over the duvet and the pockets stay at the head end.
- **Opening:** the open end of the cover, where the buttons are. It comes out of the band last and ends up at the foot end.
- **Shake:** the right gripper pinches the cover at the middle of the opening, lifts it just off the mattress, and moves it up and down twice so the duvet settles into the corners.

- **Seated:** the duvet fills all four corners of the cover, and the cover has no empty flat corner left.
- **Buttons closed:** every button on the opening is pushed through its hole. No button is left open and no hole is left empty. **Unvalidated:** the buttons are closed by the right gripper alone, with the flaps lying flat on the mattress. To be checked at the station.
- **Flat and even:** the cover lies on the mattress with no lump, no fold, and no twist, and hangs the same amount over both long sides.
- **Square on the bed:** the head end of the cover is level with the head end of the mattress, and the long edges of the cover run parallel to the long edges of the mattress.
- **Corner slipped:** a tucked corner comes out of its pocket before the cover is unrolled past it. It is tucked again at the same place.
- **Twist:** the cover is turned inside itself, so the duvet lies across the cover instead of along it. A twist is fixed by pulling the duvet back out through the opening and starting again at Step 2.
- **Just off:** lifted only far enough that the fabric no longer touches the mattress, and no higher.
- **On the floor:** any part of the duvet or the cover touches the floor. This ends the episode.
- **Back and clear:** the arm is drawn back so that no part of it is over the mattress or the fabric.

Handling standard

- Fabric stays on the mattress and slides along it. The only lift in the episode is the shake at the opening, and it is just off the mattress.
- One gripper pulls at a time. While one pulls, the other pins or holds still.
- Nothing is passed from one gripper to the other in the air.
- A corner is held all the way closed before it is pushed or pulled. A corner that slips is tucked again before anything else moves.
- Nothing leans on, pushes, drags, or lifts the bed frame, the headboard, or the mattress.

Steps

Step 1 depends on the config: the gripper on the free edge's side draws the duvet open, and in Config M the head edge is then drawn up to the band. Every other step is the same in all three.

Step 1: Open the duvet out flat

Goal: the duvet lies flat along the mattress, its head edge against the band and its long edges running along the long edges of the mattress.

1.1 Draw the top layer open

- **IF Config L:** the **right gripper** pinches the top layer of the folded duvet at the middle of its free edge and draws it along the mattress toward the foot until the duvet is open and flat.
- **IF Config M:** the **right gripper** pinches the top layer at the middle of its free edge and draws it along the mattress toward the foot until the duvet is open and flat, letting the foot end hang over the foot edge for now.
- **IF Config R:** the **left gripper** pinches the top layer at the middle of its free edge and draws it along the mattress toward the head until the duvet is open and flat and its head edge meets the band.

Check: the duvet lies open with no fold left in it. If it lies across the bed, the gripper that drew it pinches the near corner and pulls it round until it lies along the bed.

1.2 Bring the head edge to the band

- **IF Config M:** the **left gripper** pinches the duvet at the middle of its head edge and draws the whole duvet along the mattress toward the head until the head edge meets the band.
- **IF Config L or R:** nothing to do. The head edge already lies against the band.

Then, in all three:

- The **left gripper** straightens the head half and the **right gripper** straightens the foot half, so the long edges of the duvet run along the long edges of the mattress and hang the same over both.

Check: the duvet lies flat along the mattress, its head edge against the band, its long edges straight. No part of it hangs on the floor. If a corner is folded under, the gripper on that side lifts it and lays it flat.

Expected state: the duvet lies flat along the mattress with its head edge against the band. The band has not moved. Both grippers are back and clear.

Step 2: Tuck the two head corners

Goal: each head corner of the duvet sits tucked in its own pocket of the band, tip into tip.

2.1 Tuck the front head corner

- The **left gripper** pinches the front head corner of the duvet at its tip.
- Lift the corner just off the mattress and push it straight into the front pocket of the band, until the tip of the corner reaches the tip of the pocket.
- Open the gripper inside the pocket and draw it straight back out.

Check: the front head corner is tucked. The pocket sits over the corner with no slack between them. If the corner stops short, or comes out with the gripper, the **left gripper** pinches it again and pushes it in further.

2.2 Tuck the back head corner

- The **left gripper** reaches over to the back head corner of the duvet and pinches it at its tip.
- Lift it just off the mattress and push it straight into the back pocket of the band, until the tip of the corner reaches the tip of the pocket.
- Open the gripper inside the pocket and draw it straight back out.

Check: both head corners are tucked, and the band still lies straight across the head end against the headboard. The front corner has not come out while the back corner was tucked. If it has, the **left gripper** tucks it again.

Expected state: both head corners of the duvet sit in both pockets of the band. The rest of the duvet still lies flat on the mattress toward the foot. Both grippers are back and clear.

Step 3: Unroll the cover to the foot

Goal: the cover is unrolled over the whole duvet, from the pockets at the head end to the opening at the foot end, with the duvet inside it.

3.1 Pin the duvet and start the unroll

- The **right gripper** closes and pins the duvet flat at the middle of the mattress.
- The **left gripper** pinches the band at its middle, on top, and draws it along the mattress toward the foot a hand's width, so the gathered cover pays out over the duvet.
- Open, pinch the band again at its middle, and draw it another hand's width. Keep going the same way.
- Keep the fabric on the mattress. Do not lift the band up in the air.

Check: with every pull the cover moves down over the duvet a little and the pockets stay at the head end. Neither head corner has come out. If a corner slips, stop, and the **left gripper** tucks it again at the same place before the next pull. If a pull drags the duvet along instead of paying out the cover, the **right gripper** pins harder and the **left gripper** pulls again, gentler.

3.2 Swap hands past the middle

- When the band reaches the middle of the mattress, the **right gripper** opens its pin and the **left gripper** opens on the band.
- The **left gripper** closes and pins the duvet flat through the cover at the head half.
- The **right gripper** pinches the band at its middle and draws it along the mattress toward the foot a hand's width at a time, the same way, until the opening of the cover comes out of the band and reaches the foot edge of the duvet.

Check: the cover reaches from the head end to the foot end with the duvet inside it and the opening at the foot. Nothing hangs on the floor. The cover is not twisted. If the duvet lies across the cover instead of along it, the **right gripper** pulls the duvet back out through the opening and the episode starts again at Step 2.

3.3 Push the foot corners in

- The **left gripper** opens its pin and comes back and clear.
- The **right gripper** pinches the front foot corner of the duvet and pushes it into the front near corner of the cover inside the opening until it fills the corner, then does the same with the back foot corner.

Check: the duvet reaches all four corners of the cover and no part of it hangs out of the opening. If a foot corner sits short, the **right gripper** pushes it in until it fills the corner.

Expected state: the duvet is inside the cover, head end to foot end. The opening is at the foot end and still open. Both grippers are back and clear.

Step 4: Shake the opening to seat the duvet

Goal: the duvet is seated in all four corners of the cover, with no empty corner and no lump.

4.1 Shake it twice

- The **right gripper** pinches the cover at the middle of the opening, both flaps together, so nothing falls out.
- Lift the opening just off the mattress, no higher than the mattress top, and move it up and down twice, so the duvet settles along inside the cover.
- Lower it back onto the mattress and open. Keep the head end on the mattress the whole time. Do not lift the whole load up in the air.

Check: the head end of the cover is full, not flat and empty, and both foot corners are still full. If a corner is empty, the **right gripper** shakes twice more. Do no more than two extra shakes.

4.2 Smooth it down

- The **left gripper** runs its closed face along the top of the cover over the head half, from the head end to the middle, flat and gentle, pushing any lump toward the nearest edge.
- The **right gripper** does the same over the foot half, from the middle to the foot end.

Check: the duvet is seated in all four corners. The cover has no big lump and no fold caught inside it. If the duvet lies across the cover instead of along it, the **right gripper** pulls the duvet back out through the opening and the episode starts again at Step 2.

Expected state: the duvet is seated in the cover. The opening is at the foot end, still open. Both grippers are back and clear.

Step 5: Close the buttons

Goal: every button on the opening is closed, and the opening lies flat.

5.1 Line the opening up

- The **right gripper** pinches the top flap of the opening at its middle and lays it over the bottom flap, so the buttons line up with their holes.
- Press the opening flat on the mattress with the closed face of the gripper.

Check: each button sits next to its own hole and no flap is folded under. If the flaps are out of line, the **right gripper** lifts the top flap and lays it over again.

5.2 Close each button in order

- Start at the button nearest the front edge and work toward the back edge.
- The **right gripper** pinches the button, pushes it through its hole with the hole flap lying flat on the mattress, and opens.
- Do the next button the same way, and keep going to the last one. Do not skip a button and do not go back.

Check: every button is through its own hole. No button is left open and no hole is left empty. If a button will not go through, the **right gripper** presses the hole flap flat with its closed face and pushes the button again. Try no more than three times on one button.

5.3 Look along the closed opening

- The **right gripper** runs its closed face along the closed opening from the front edge to the back edge, flat and gentle, pressing the opening flat against the mattress.

Check: the opening lies flat and closed, with no gap between the buttons and no corner of the duvet poking out. If a corner pokes out, the **right gripper** opens the nearest button, pushes the corner back in, and closes the button again.

Expected state: the cover is closed with all its buttons. The duvet is inside it. Both grippers are back and clear.

Step 6: Spread it on the bed

Goal: the cover lies square and flat over the whole mattress, hanging the same amount over both long sides.

6.1 Square it to the mattress

- The **left gripper** pinches the cover at the middle of its head end and draws it up the bed until the head end of the cover is level with the head end of the mattress.
- The **left gripper** straightens the head half so the long edges of the cover run along the long edges of the mattress.

Check: the head end of the cover is level with the head end of the mattress, and the cover is not turned across the bed. If it sits low or turned, the **left gripper** draws it up and round again.

6.2 Even the two sides

- The **right gripper** pinches the cover at the middle of its far long edge and draws it out until it hangs over the far side.
- The **right gripper** pinches the cover at the middle of its near long edge and draws it out until it hangs over the near side the same amount.
- Look along the bed from the side to see that both hangs look the same.

Check: the cover hangs the same amount over both long sides. If one side hangs much longer, the **right gripper** draws the cover across toward the short side until the two look the same.

6.3 Smooth it flat

- The **left gripper** runs its closed face over the top of the cover from the head end to the middle, pushing every fold and lump toward the nearest edge.
- The **right gripper** runs its closed face over the top of the cover from the middle to the foot end the same way.
- Do not lift the cover off the mattress to do it.

Check: the cover lies flat and even, with no fold, no lump, and no twist. The whole mattress is covered. If a fold is left, the gripper on that side smooths that spot again.

Expected state: the duvet is inside the closed cover, and the cover lies flat, square, and even over the whole mattress. Both grippers are back and clear.

Step 7: End the episode

Goal: the duvet is in its cover, spread on the bed, and both arms are safely home.

- Confirm the duvet is inside the cover, seated in all four corners, with no part of it out of the opening.
- Confirm the cover is closed at every button, right side out, and not twisted.
- Confirm the head end of the cover is level with the head end of the mattress, the cover lies flat, and both sides hang the same.
- Confirm no part of the duvet or the cover touches the floor, the mattress and the frame stand where they started, and the sheet under the cover is not pulled loose.
- Confirm the base has not moved: it is still square to the bed and still locked.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded. The base stays parked and locked through the reset, and is only pushed away by hand once the bed is set for the next episode.

1. Open every button on the cover opening.

2. Pull the duvet out of the cover through the opening.
3. Shake the duvet out, fold it in half once across its length, and lay it at the duvet spot the next episode's config puts it: the head half against the band for Config L, centered for Config M, the foot half for Config R, with its free edges toward the foot for Config L and M and toward the head for Config R.
4. Turn the cover right side out if it came out inside out. Gather it by hand from its opening back to its closed end into a band no wider than a hand, and lay the band across the head end of the mattress against the headboard, pockets facing the foot, buttons open.
5. Smooth the fitted sheet on the mattress and tuck it back in at every corner.
6. Check the mattress holds nothing but the folded duvet and the band.
7. Replace the cover if a button is missing, a button hole is torn, or the fabric is torn.
8. Replace the duvet if it is torn, damp, or too big for the cover.
9. Replace either one if it touched the floor during the episode. Send it to the wash.
10. Check the floor strip along the side of the bed is clear and dry.
11. Check the base is still square, centered, and locked, and push it firmly once by hand to confirm it does not roll. If it has moved, park it again by the Base positioning steps.
12. Run the Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the duvet spot: the violations below were written for Config L (folded duvet in the head half, drawn open by the right gripper). The arm-role cues in them will be rewritten later to cover all three configs; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the grippers do does not match the config on the bed. The gripper on the far side reaches across to draw the duvet open; the head edge is drawn up to the band in a

config where it already lies there, or is not drawn up in Config M; the folded duvet does not lie where that config puts it; or the wrong IF line is followed in Step 1.

- **SOP rule broken:** Steps 1.1 and 1.2, the same-side rule. The gripper on the free edge's side draws the duvet open, and the IF line followed is the one for the config on the bed.
- **Coaching note:** look where the folded duvet lies before anything is touched, then follow that config's IF lines through Step 1.

Violation: Base moved during the episode

- **Visible cue:** the bed shifts in the frame, the mattress edge changes angle in the frame, or the base rolls, creeps, or turns at any time after recording starts.
- **SOP rule broken:** Step 7 and the Base positioning rule it confirms, the base is parked and locked before recording and does not move at all during the episode.
- **Coaching note:** park it, lock it, push-test it. If the park is wrong, fix it before recording, never during.

Violation: Fabric touched the floor

- **Visible cue:** any part of the duvet or the cover falls off the bed and touches the floor at any time.
- **SOP rule broken:** Steps 1 to 6, nothing ever touches the floor.
- **Coaching note:** keep the load on the mattress. If it is sliding off, pinch it lower down, not further out.

Violation: Lifted the load up in the air

- **Visible cue:** a gripper lifts the duvet, the band, or the cover clear of the mattress and holds it up, instead of sliding it along the mattress.
- **SOP rule broken:** Steps 1, 3, and 4.1, all the work stays on the mattress, and only the opening lifts, just off, and only for the shake.
- **Coaching note:** slide it, do not carry it. The only lift in this task is the shake at the opening.

Violation: Head corners not tucked first

- **Visible cue:** the band is pulled toward the foot before both head corners are in their pockets, or only one head corner is tucked before the unroll starts.
- **SOP rule broken:** Step 2, both head corners are tucked, tip into tip, before anything is unrolled.
- **Coaching note:** two corners first, every time. A cover unrolled without its corners comes out crooked.

Violation: Corner slipped and not retucked

- **Visible cue:** a tucked corner comes out of its pocket during Step 3 and the pulling carries on without the left gripper tucking it again.
- **SOP rule broken:** Steps 2 and 3.1, a tucked corner stays in its pocket until the cover is unrolled past it, and a corner that slips is tucked again at the same place before the next pull.
- **Coaching note:** stop the moment it slips. Two seconds now saves a full restart later.

Violation: Cover put on twisted

- **Visible cue:** the duvet ends up lying across the cover instead of along it, or the cover is turned inside itself, so one corner of the duvet is nowhere near a corner of the cover.
- **SOP rule broken:** Steps 3.2 and 4.2, the cover unrolls straight down the duvet, and a twist is fixed by pulling the duvet out through the opening and starting again at Step 2.
- **Coaching note:** check the band is straight before the first pull. A twist cannot be smoothed out from the top.

Violation: Duvet opened crooked or hanging off

- **Visible cue:** the duvet is left lying across the bed, with a fold still in it, with a corner folded under, or with its head edge short of the band when the tucking starts.
- **SOP rule broken:** Step 1, the duvet lies flat along the mattress with its head edge against the band and its long edges straight before a corner is tucked.
- **Coaching note:** flat, straight, and up to the band. The corners only go in cleanly from a flat duvet.

Violation: Duvet not seated in the corners

- **Visible cue:** at the end of Step 4, a corner of the cover is flat and empty, or the duvet ends short of one end.
- **SOP rule broken:** Steps 3.3 and 4.1, the duvet fills all four corners of the cover.
- **Coaching note:** push the foot corners in, shake twice, then look at all four corners. An empty corner shows straight away.

Violation: Shaken too hard or too high

- **Visible cue:** the opening is lifted well above the mattress top, the load is swung, or the shake throws the fabric off the bed.
- **SOP rule broken:** Step 4.1, lift the opening just off the mattress, no higher, and move it up and down twice.
- **Coaching note:** small and twice. A big swing puts the duvet on the floor.

Violation: Buttons left open or skipped

- **Visible cue:** at the end of Step 5 a button is still open, a hole is empty, or the buttons are done out of order with one missed in the middle.
- **SOP rule broken:** Step 5.2, every button is closed, in order, from the front edge to the back edge.
- **Coaching note:** front to back, no skipping. Run back along the line before calling it closed.

Violation: Duvet poking out of the opening

- **Visible cue:** a corner or an edge of the duvet sticks out through the closed opening at the end of Step 5.
- **SOP rule broken:** Step 5.3, no corner of the duvet pokes out of the closed opening.
- **Coaching note:** push it in, then close the button. Never close a button over a corner.

Violation: Leaned on or moved the bed

- **Visible cue:** a gripper, wrist, or forearm rests on the bed frame or the headboard, the mattress slides on the frame, or the fitted sheet is pulled loose.
- **SOP rule broken:** Steps 1 to 6, nothing leans on, pushes, drags, or lifts the bed frame, the headboard, or the mattress.
- **Coaching note:** the arm holds itself up. If the bed moves under a pull, the pull is from the wrong place.

Violation: Pulled without a pin or both pulled at once

- **Visible cue:** the band is drawn along the mattress while no gripper pins the duvet, both grippers pull at the same time, or the same gripper keeps pulling after the band has crossed the middle.
- **SOP rule broken:** Steps 3.1 and 3.2, one gripper pins the duvet while the other pulls, and the hands swap when the band reaches the middle.
- **Coaching note:** one pins, one pulls, then swap at the middle. Two pulls at once just drags the whole duvet.

Violation: Done in the wrong order

- **Visible cue:** the buttons are closed before the foot corners are pushed in, the cover is spread before the buttons are closed, or the shake happens before the cover reaches the foot.
- **SOP rule broken:** Steps 1 to 6, open, tuck the head corners, unroll, push the foot corners in, shake, close the buttons, spread.
- **Coaching note:** the order is the task. Every step out of order has to be undone.

Violation: Spread crooked or uneven

- **Visible cue:** at the end of Step 6 the head end of the cover sits well short of the head end of the mattress, the cover runs across the bed, or one side hangs far longer than the other.
- **SOP rule broken:** Steps 6.1 and 6.2, the cover is square to the mattress and hangs the same amount over both sides.
- **Coaching note:** head end first, then even the sides. Look along the bed from the side before stopping.

Violation: Folds and lumps left in

- **Visible cue:** at the end of Step 6 the cover shows a big fold, a lump, or a ridge across the top.
- **SOP rule broken:** Steps 4.2 and 6.3, each gripper smooths its own half flat, from the head end to the foot end.
- **Coaching note:** push every lump out to the nearest edge. Lumps do not settle on their own.

Violation: Wrong arm used

- **Visible cue:** the right gripper tucks a head corner or squares the head end, the left gripper pushes a foot corner in, shakes the opening, closes a button, or evens a long edge, or a gripper works the other half of the mattress when smoothing.
- **SOP rule broken:** Steps 2, 3.3, 4, 5, and 6, the left gripper owns the head end and the head half, the right gripper owns the foot end and the foot half.
- **Coaching note:** left arm at the head, right arm at the foot. Each arm smooths its own half.

Violation: Something dropped or knocked off the bed

- **Visible cue:** the duvet or the cover slips out of a gripper and falls off the side of the bed, or is knocked off by an arm going past.
- **SOP rule broken:** Steps 1 to 6, one load moves at a time, and each one is put down under control.
- **Coaching note:** close on the fabric all the way before pulling, and put it down before letting go.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails and the episode carries on with no retry: a corner sits short, a corner is empty after the shake, or a button is left open.
- **SOP rule broken:** Steps 1 to 6, do the retry written under each failed check before moving on.
- **Coaching note:** a check is only worth doing if the retry is done. Fix it where it happened.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the duvet part out of the cover, a button open, the cover crooked on the bed, fabric on the floor, or an arm away from home.
- **SOP rule broken:** Step 7 (confirm the duvet, the cover, the spread, the floor, the bed, and the locked base; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see the head corners, the opening, or the buttons**, so nobody can tell if the corners were tucked or the buttons were closed.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake that releases on its own, or a caster that seizes so the base cannot be parked square.
- **Defective object:** a cover with a missing button, a torn button hole, a torn seam, or an opening that will not lie flat when closed; a duvet that is torn, damp, lumpy inside, or too big to go into the cover at all; a duvet and a cover that are not the same size, so the corners cannot be tucked. Replace it before the next episode.
- **Bed fault:** a mattress that slides on its frame under a correct pull, a frame that rocks, or a fitted sheet that comes off its corners under a correct pull, so the mattress cannot be worked on.
- **Band fault:** a band that was gathered inside out or laid with its pockets facing the headboard before recording started.
- **Reach fault:** a place turns out to be too far for its arm with the base parked correctly, so a head corner, a foot corner, the band, or the folded duvet cannot be reached without extending the arm to a limit.

Annotation subtasks (from SOP)

1. Draw the duvet open flat along the mattress
2. Bring the head edge of the duvet to the band
3. Tuck the front head corner into its pocket
4. Tuck the back head corner into its pocket
5. Pin the duvet and unroll the band through the head half
6. Swap hands and unroll the band through the foot half
7. Push the foot corners into the cover

8. Shake the opening to seat the duvet
9. Smooth the duvet down inside the cover
10. Line the opening up
11. Close each button in order
12. Square the cover to the mattress
13. Even the two sides
14. Smooth the cover flat
15. Return both arms home and end the episode